

Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 11



By- Ravindra Sir

Recap of Previous Lecture



Topic

Regular

Topic

Topic

Topics to be Covered



Topic

Regular

Topic

Topic

- If L_1 is Reg, L_2 is Reg $\Rightarrow L_1 \cup L_2$ is Reg
- If $L_1 \cup L_2$ is Reg then L_1, L_2 may or may not be Reg -
- If $L_1 \cap L_2$ is Reg " " " " -
- If $L_1 \cdot L_2$ is Reg " " " "
- If $\bar{L}$ is Reg, then L is Reg
- If L^R is Reg, then L is Reg

Union:-

$$\frac{L_1}{R} \quad \frac{L_2}{R}$$

$$a^*$$

$$b^*$$

$$L_1 \cup L_2 = a^* + b^* \rightarrow \text{Reg}$$

$$\frac{L_1}{R}$$

$$\frac{L_2}{NR}$$

$$(a+b)^*$$

$$a^n b^n$$

$$L_1 \cup L_2 = (a+b)^* \rightarrow \text{Reg}$$

$$\frac{L_1}{NR}$$

$$\frac{L_2}{NR}$$

$$a^n b^n$$

$$a^n b^n$$

$$L_1 \cup L_2 = (a+b)^* \rightarrow \text{Reg}$$



intersection:-



$$\begin{array}{l} \underline{L_1} \\ R \\ a^* \end{array}$$

$$\begin{array}{l} \underline{L_2} \\ R \\ b^* \end{array}$$

$$L_1 \cap L_2 = \{\epsilon\} \rightarrow \text{Reg}$$

$$\begin{array}{l} \underline{L_1} \\ R \\ \emptyset \end{array}$$

$$\begin{array}{l} \underline{L_2} \\ NR \\ a^n b^n \end{array}$$

$$L_1 \cap L_2 = \emptyset \rightarrow \text{Reg}$$

$$\begin{array}{l} \underline{L_1} \\ NR \\ a^n b^n \end{array}$$

$$\begin{array}{l} \underline{L_2} \\ NR \\ a^n b^n \end{array}$$

$$L_1 \cap L_2 = \emptyset \rightarrow \text{Reg}$$

- If L_1 is regular $\iff \bar{L}_1$ is regular
- If L_1 is not regular $\iff \bar{L}_1$ is not regular
- $(L_1, \bar{L}_1)$ are both regular or both are non regular
- L_1 is reg and L_2 is reg $\Rightarrow L_1 \cup L_2$ is reg ($p \rightarrow q$)
- If $L_1 \cup L_2$ is non Reg $\Rightarrow L$ is NR or L_2 is NR ($\sim q \rightarrow \sim p$)

	Reg	DCFL	CSL	Rec	
$\bar{L}$	✓	✓	✓	✓	
→ L and $\bar{L}$	are both regular	Δ	L and $\bar{L}$	are both non Reg	
→	"	"	DCFL	"	" not DCFL
→	"	"	CSL	"	" not CSL
→	"	"	Rec	"	" not Rec
→	"				

→ If L^* is reg $\Rightarrow L$ may or may not be Reg

$$L = \{a\} \quad L^* = a^*$$

Reg ✓

$$L_1 = \{a^n / n \geq 1\} = \{\underline{a}, aaaa, a^9, \dots\}$$

Non Reg ✓

$$L_1^* = a^* \checkmark$$

$$L = \{a^{2^n} / n \geq 1\}$$

$$L = \{a^2, a^4, a^8, a^{16}, \dots\}$$

$$\underline{\underline{(aa)^*}}$$

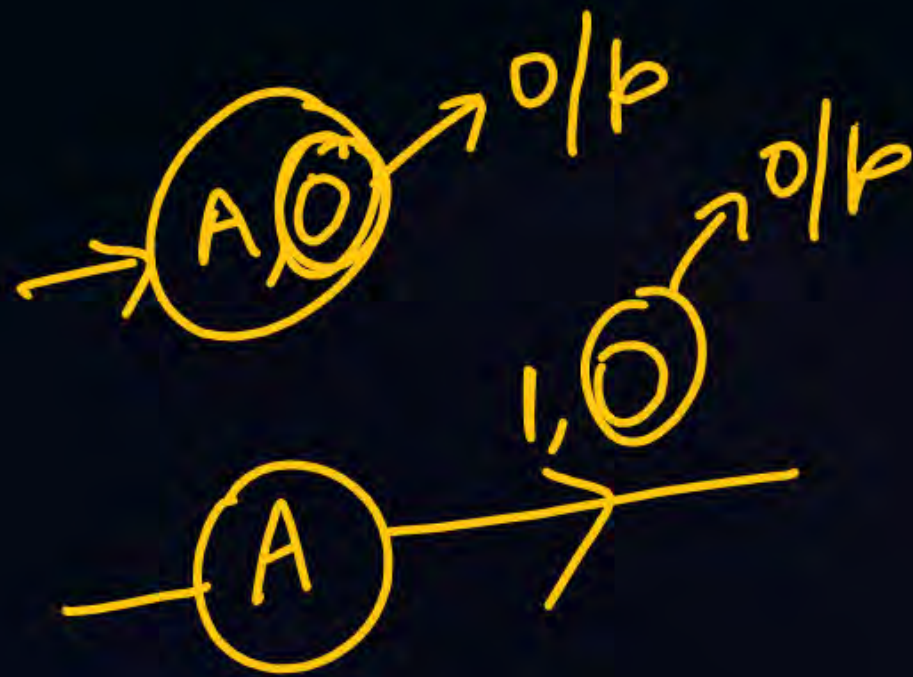
→ If a problem has an algorithm, then the problem is called decidable.

	Reg	CFL	CSL	Rec	RE
$w \in L$ membership	D	D	D	D	UD
$L \stackrel{?}{=} \emptyset$ Emptiness	D	D	UD	UD	UD
L finite?	D	UD	UD	UD	UD
equivalence	D	UD	UD	UD	UD
$L_1 = L_2$	D	UD	UD	UD	UD
$L = R$ Regularity	D	UD	UD	UD	UD
$L \rightarrow$ ambiguous ambiguity	D	UD	UD	UD	UD
$L \stackrel{?}{=} \Sigma^*$ Completeness	D	UD	UD	UD	UD
$L_1 \cap L_2 = \emptyset$ Disjointness	D				

Applications of FA:-

- 1) lexical analyzer (C/D)
- 2) Text editor → Search
auto suggestion
GREP → unit
- 3) seq circuits (digital)

mode / mealy:-
↓
DFA



Construct a Moore m/c that takes set of all strings over $\{a, b\}$ and prints '1' as o/p for every occurrence of 'ab' as a substring. (ending with ab) otherwise '0'

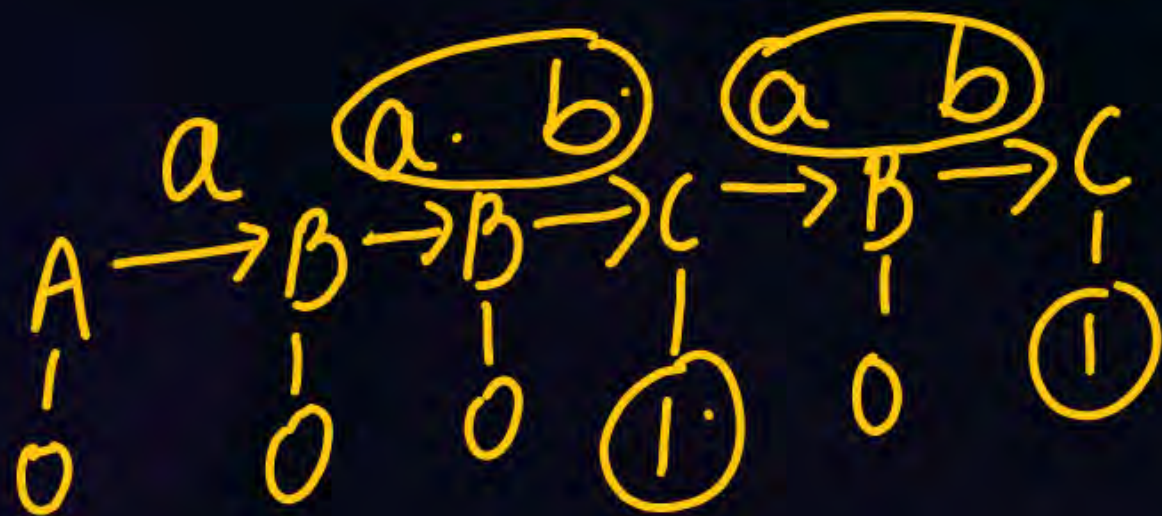
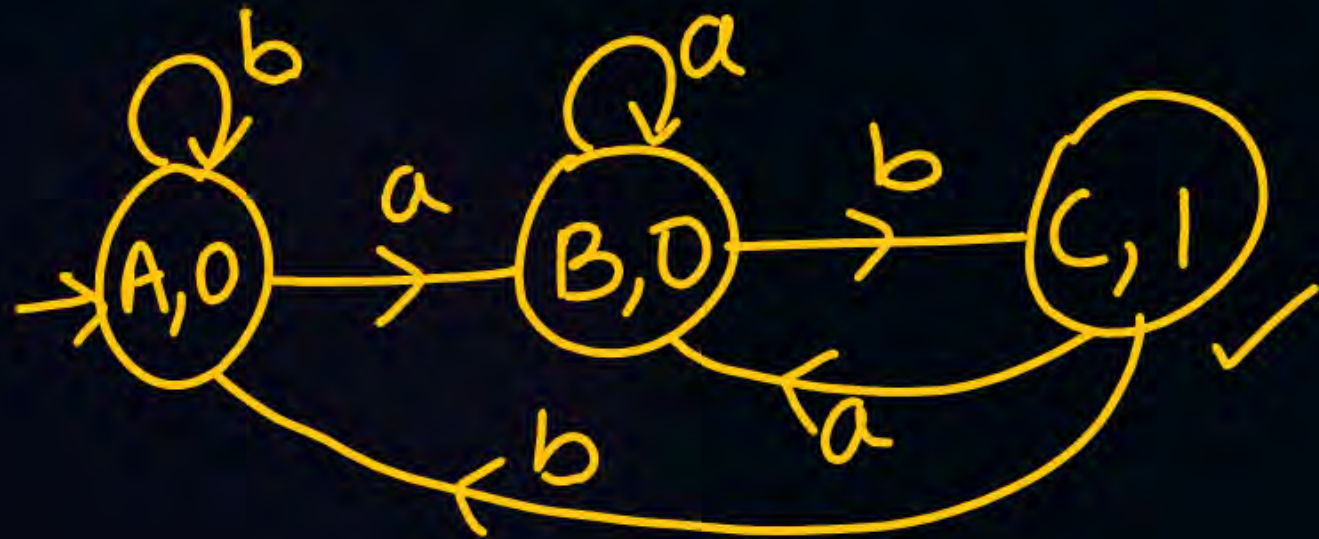
mealy and moore $\rightarrow$ no final state

mealy and moore $\rightarrow$ transducer

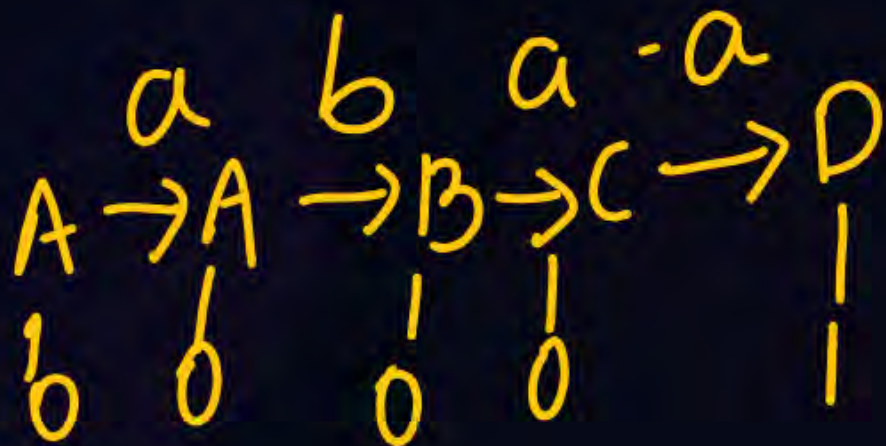
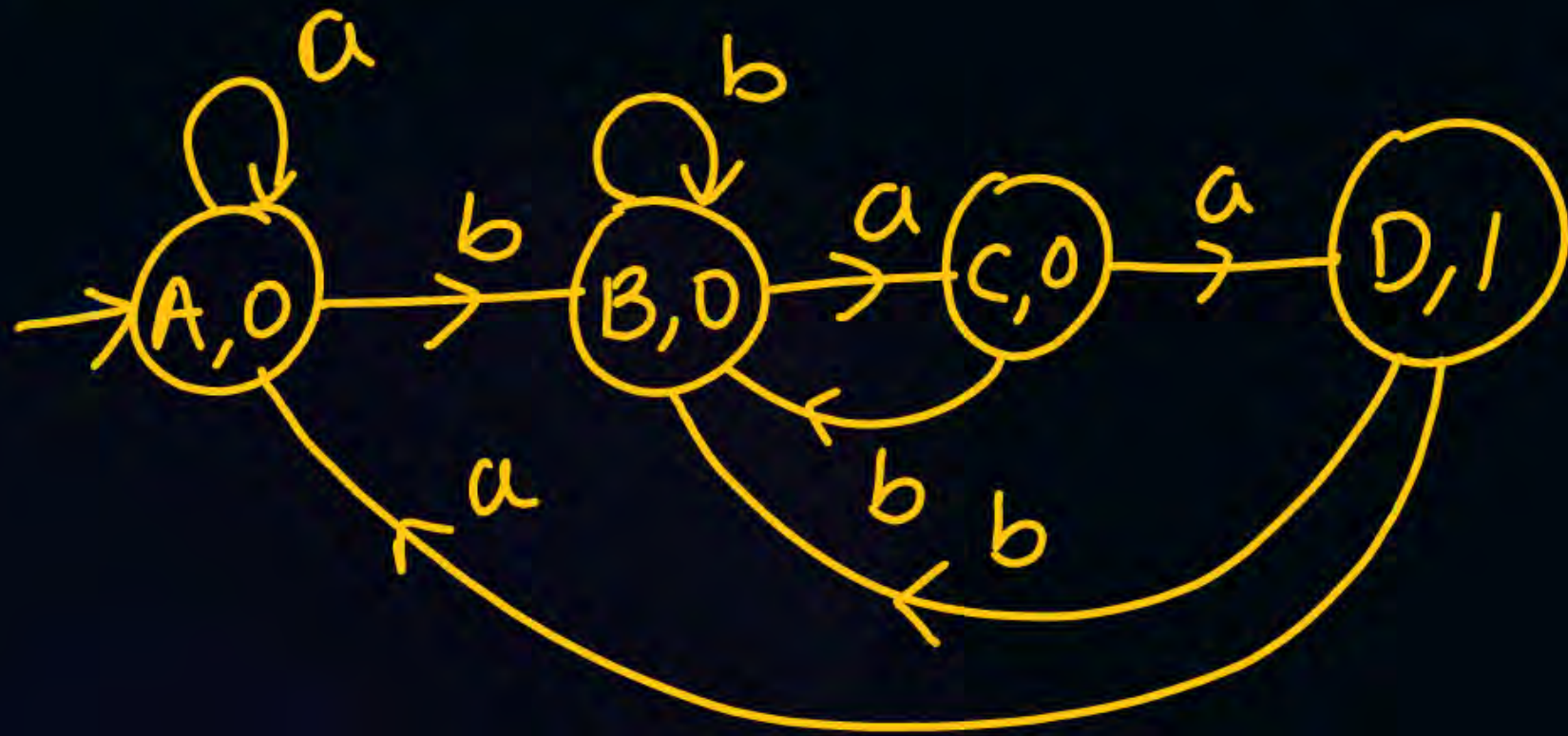
$\hookrightarrow \underline{\underline{1/p}} \rightarrow \underline{\underline{0/p}}$



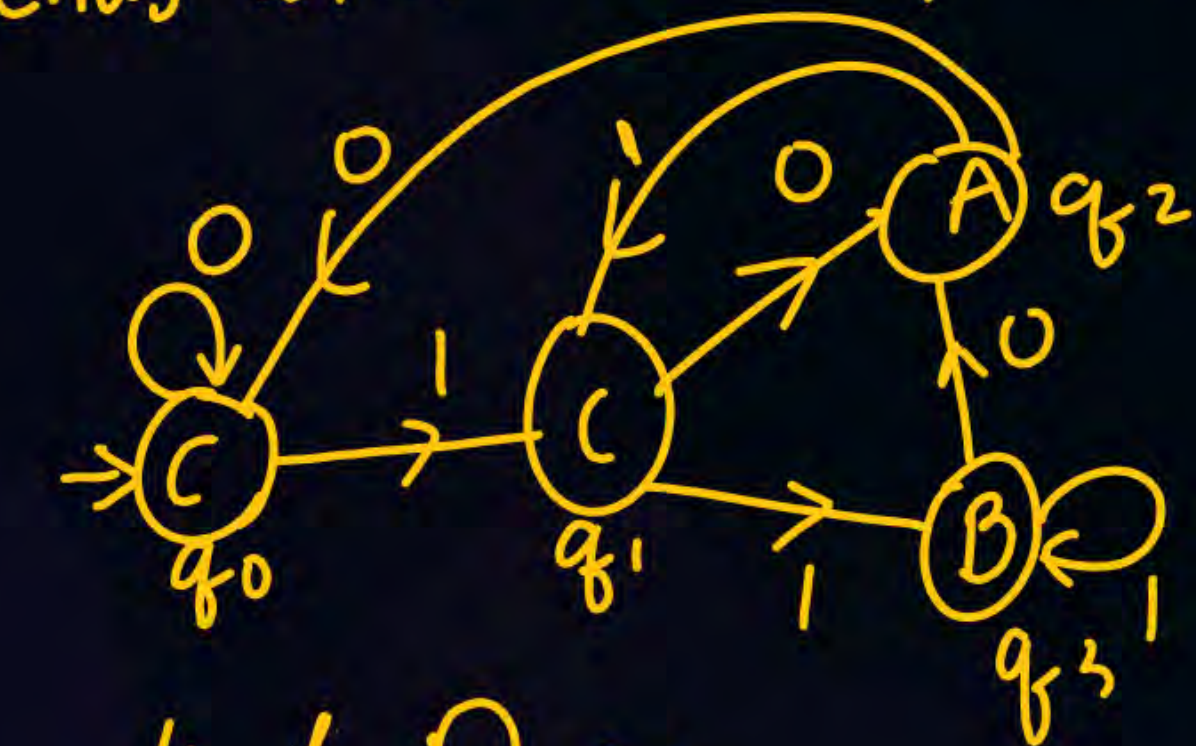
Construct a finite m/c that takes set of all strings over $\{a, b\}$ and prints '1' as o/p for every occurrence of "ab" as a substring. (ending with ab) otherwise '0'



Construct a finite m/c that prints '1' for every occurrence of 'baa', otherwise '0' (ending with baa)



→ Construct a finite m/c that takes set of all strings over $\{0,1\}$ and produces 'A' as o/p if i/p ends with '10' & produces B as o/p if i/p ends with '11' & produces 'C' otherwise.



→ Construct a Moore m/c that takes binary number as i/p and produces residue modulo '3' as o/p.

$$\underline{\underline{o/p\ 3}} \rightarrow \underline{\underline{0, 1, 2}}$$

	0	1	o/p
→ q ₀	q ₀	q ₁	0
q ₁	q ₂	q ₀	1
q ₂	q ₁	q ₂	2

Construct a Moore m/c that takes base 4 number as i/p and produces residue modulo 5 as o/p.

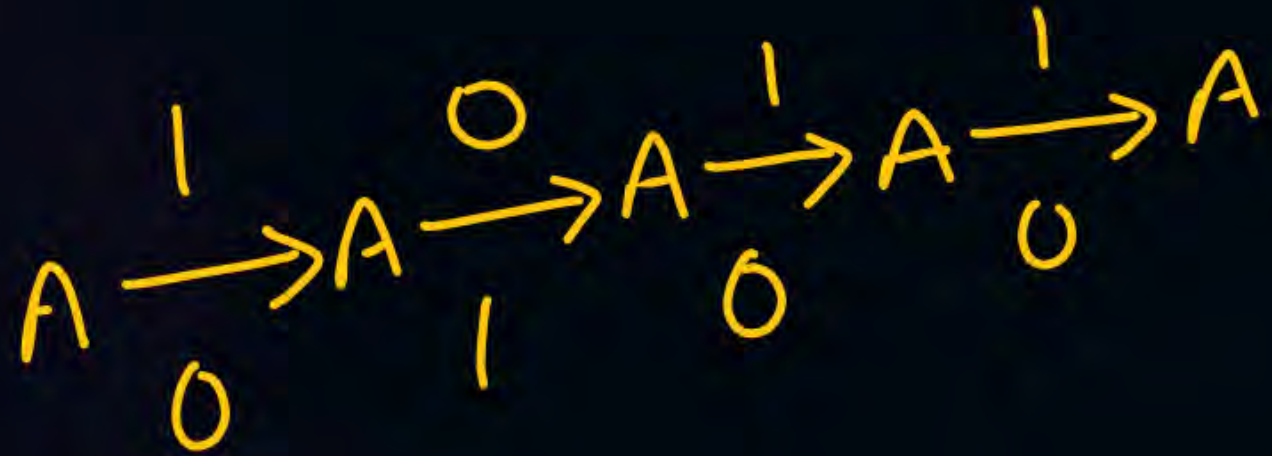
	0	1	2	3	o/p
→ q ₀	q ₀	q ₁	q ₂	q ₃	0
q ₁	q ₄	q ₀	q ₁	q ₂	1
q ₂	q ₃	q ₄	q ₀	q ₁	2
q ₃	q ₂	q ₃	q ₄	q ₀	3
q ₄	q ₁	q ₂	q ₃	q ₄	4

moore:-

o/p is associated with state

mealy:-

Construct a mealy m/c that takes binary numbers as i/p
and produces 1's complement as o/p



Ques:- Construct a mealy m/c for 2's Complement, assuming that i/p is read from LSB to MSB and carry is discarded.

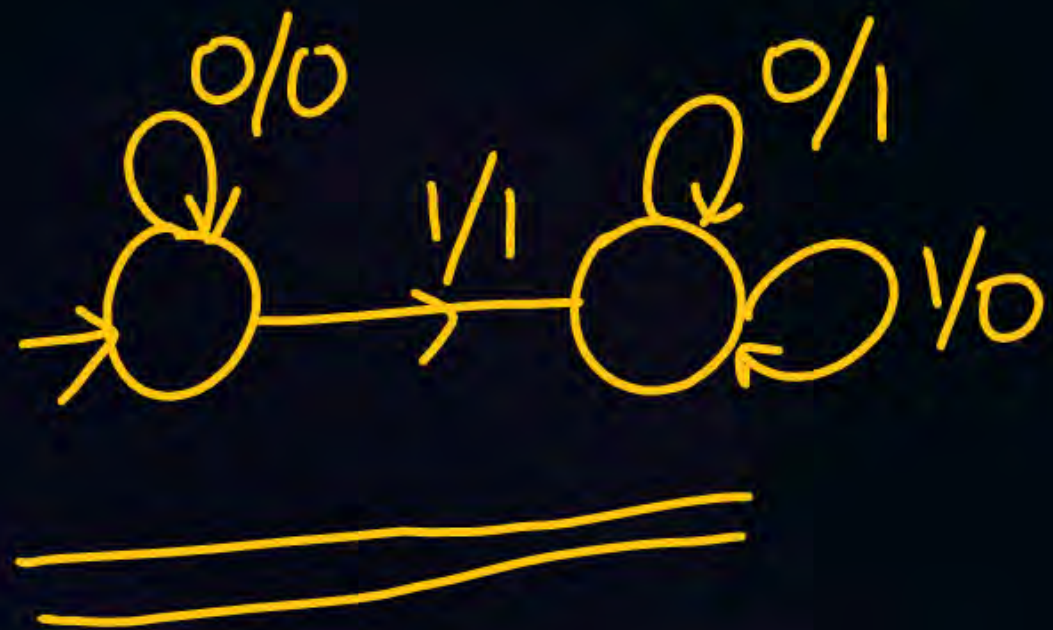
$$\begin{array}{r} 1010\textcircled{1}00 \\ \hline \end{array}$$

$$\begin{array}{r} 0101100 \\ \hline \end{array}$$

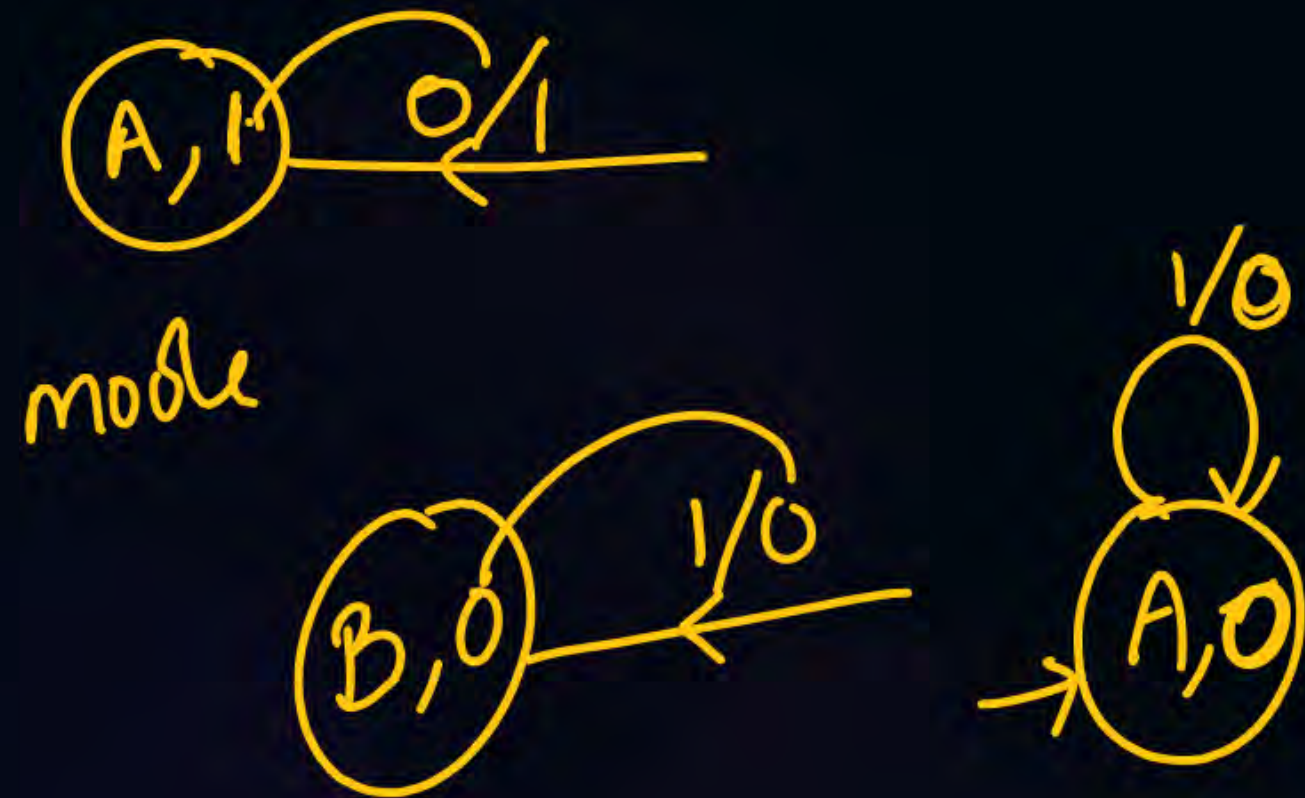
$$\begin{array}{r} 01010\textcircled{1}00 \\ \hline 10101100 \\ \hline \end{array}$$

$$\begin{array}{r} 0101010\textcircled{1} \\ \hline 1010101 \rightarrow 2's \text{ Comp} \\ \hline \end{array}$$

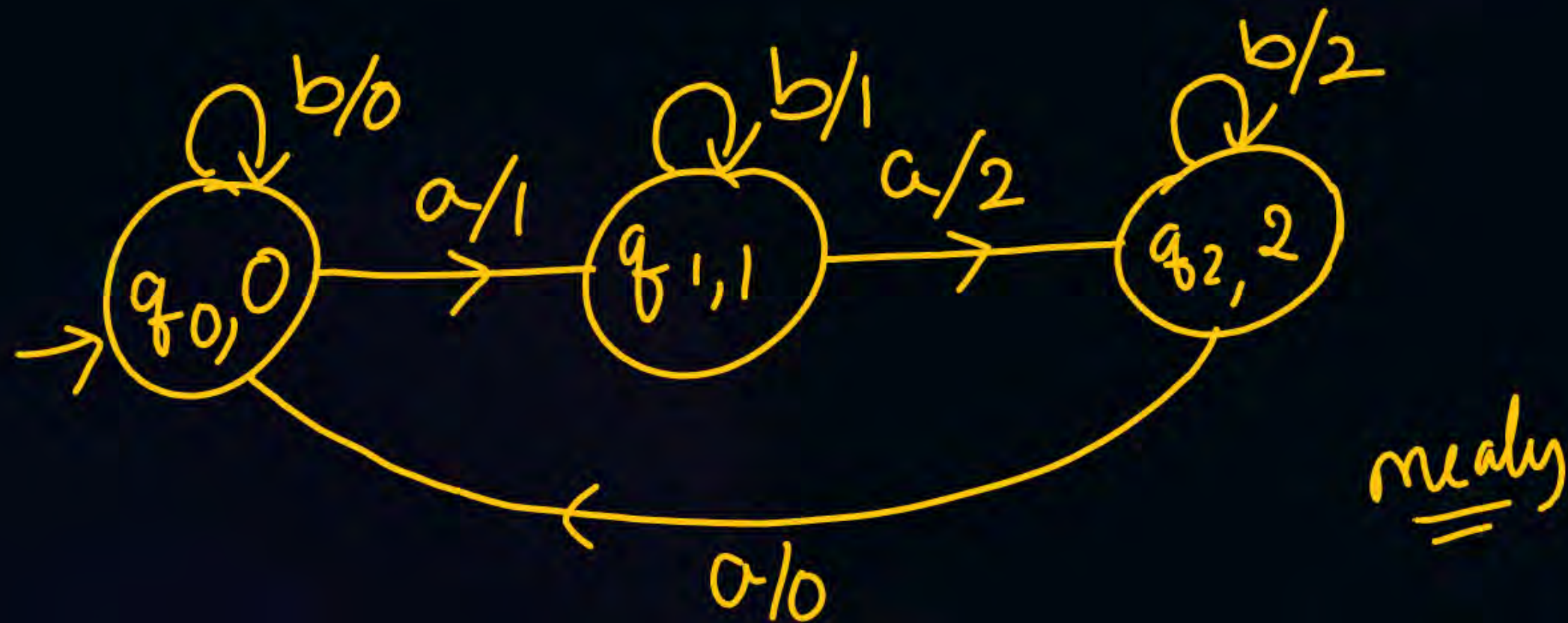
Ques:- Construct a mealy m/c for 2's Complement, assuming that i/p is read from LSB to MSB and carry is discarded.

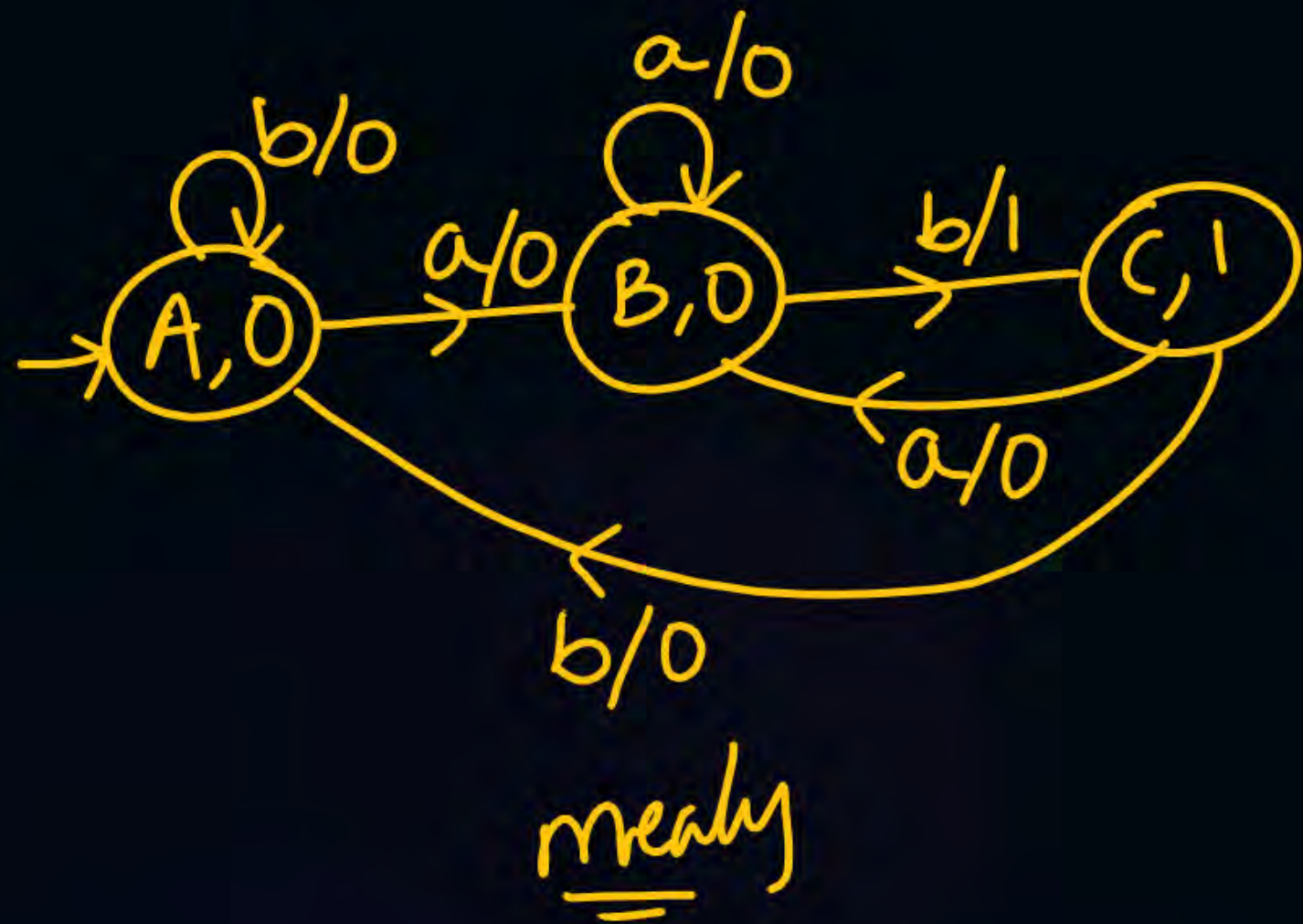


mode $\rightarrow$ mealy

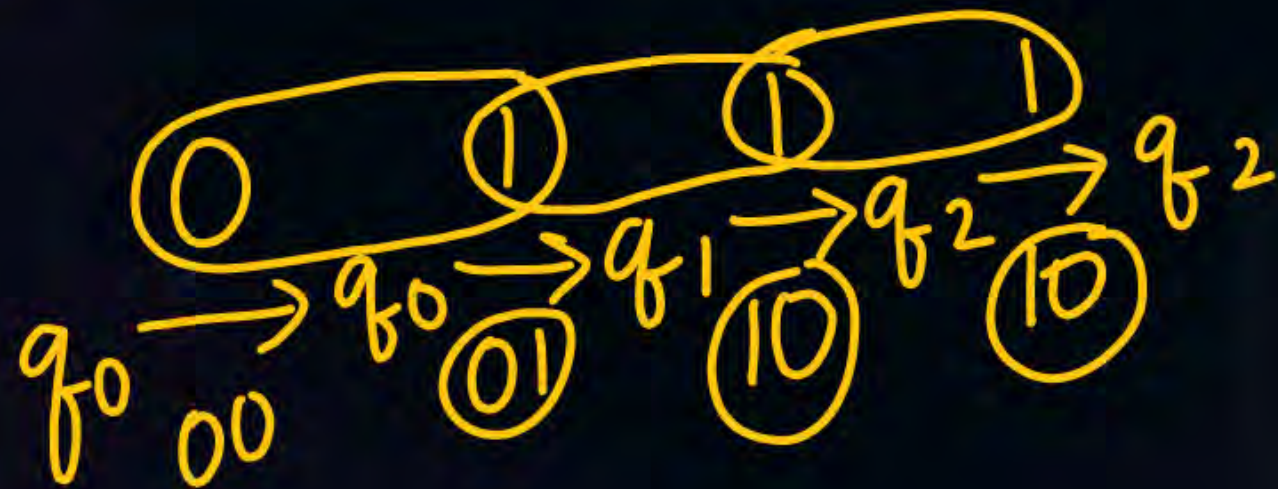
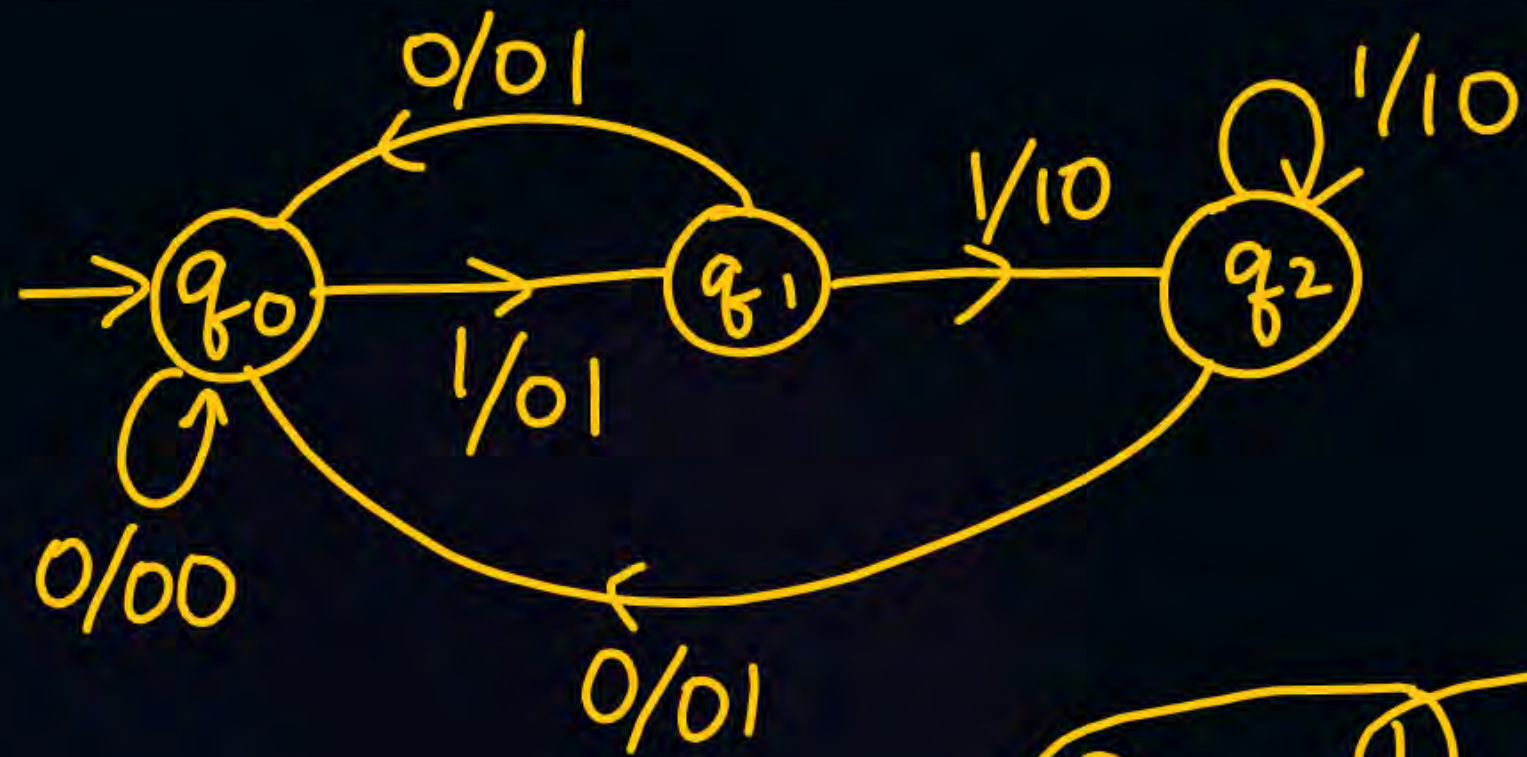


Construct a finite m/c for counting no of a's mod 3

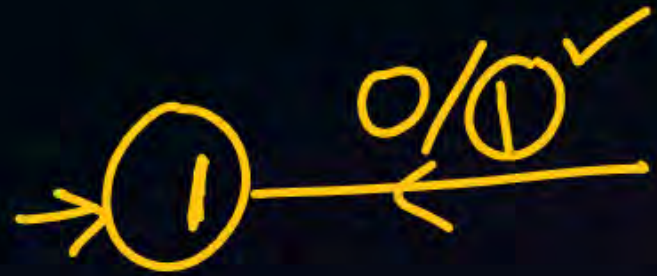




What is the o/p produced by following m/c?

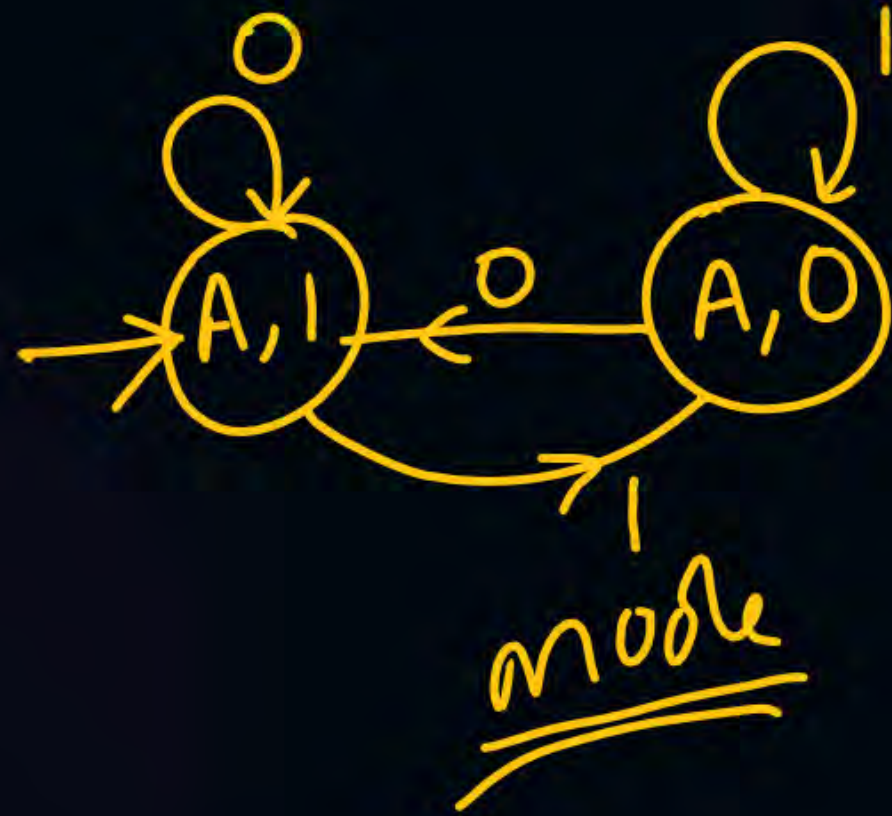
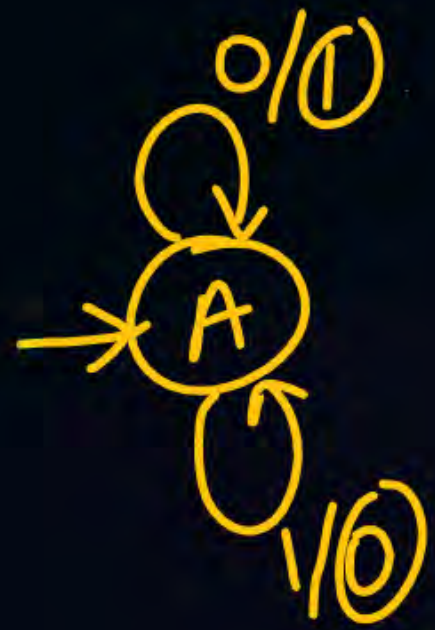


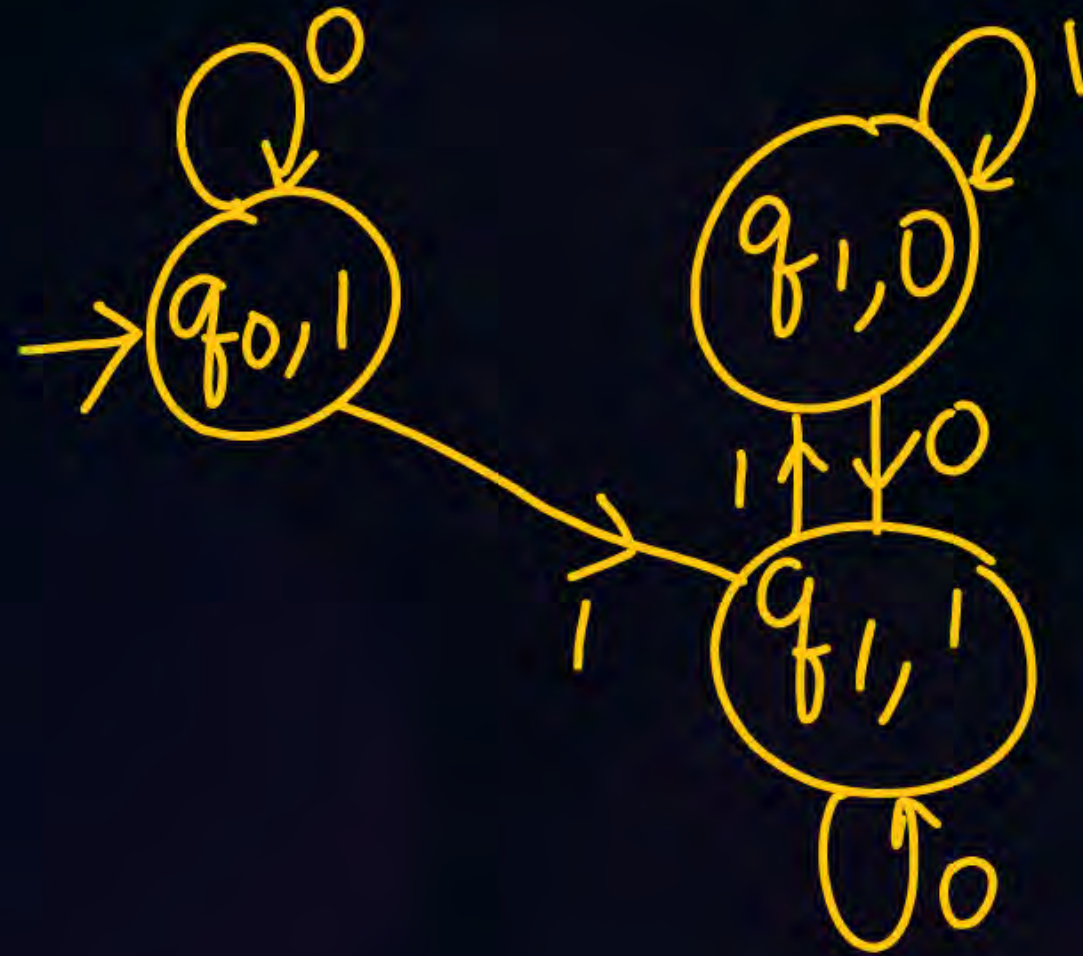
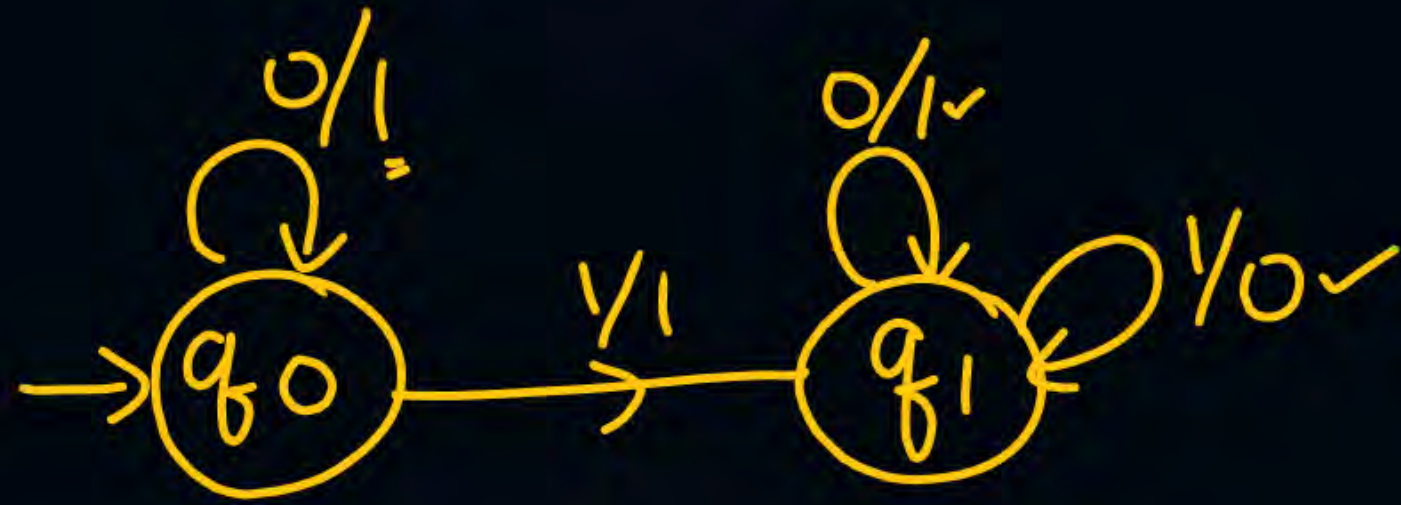
mealy to moore:-



$A, 0$

$A, 1$





mode machine



THANK - YOU